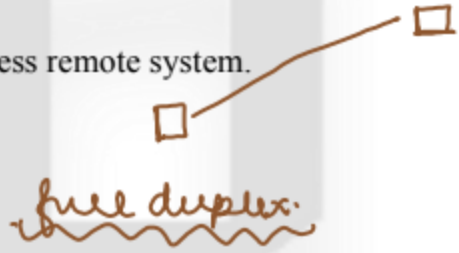
Application Layer: Domain Name System, World Wide Web and HyperText Transfer Protocol, Electronic mail, File Transfer Protocol, Remote login, Network management, Data compression, Cryptography - basic concepts.

TELNET (TErminaL NETwork)

- Command line tool (Text-Oriented) to access remote system.
- Teletype Network.
- Developed in 1969.
- Client-Server Protocol (Bidirectional).
- Uses port 23



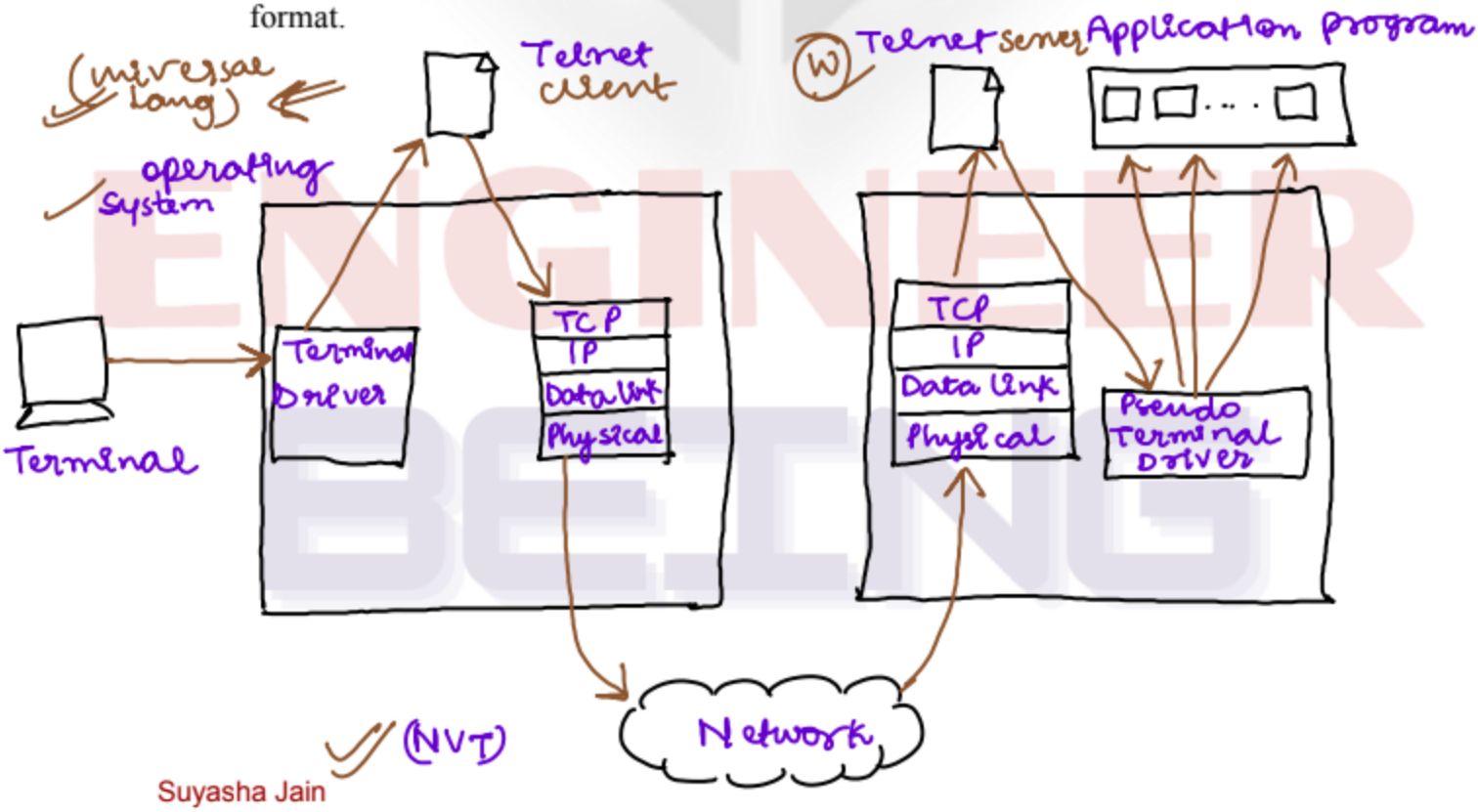
WHY TELNET IS USED?

- Command line tool to access remote computer.
- Edit files, run programs.
- To check open ports.
- Configure network devices like routers, switches etc.

TELNET client converts the universal language information

NVT Network virtual terminal sets a standard for both the systems to communicate and ensures how the data is transferred irrespective of OS

TELNET SERVER Converts from universal language to system understandable format.



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CLI

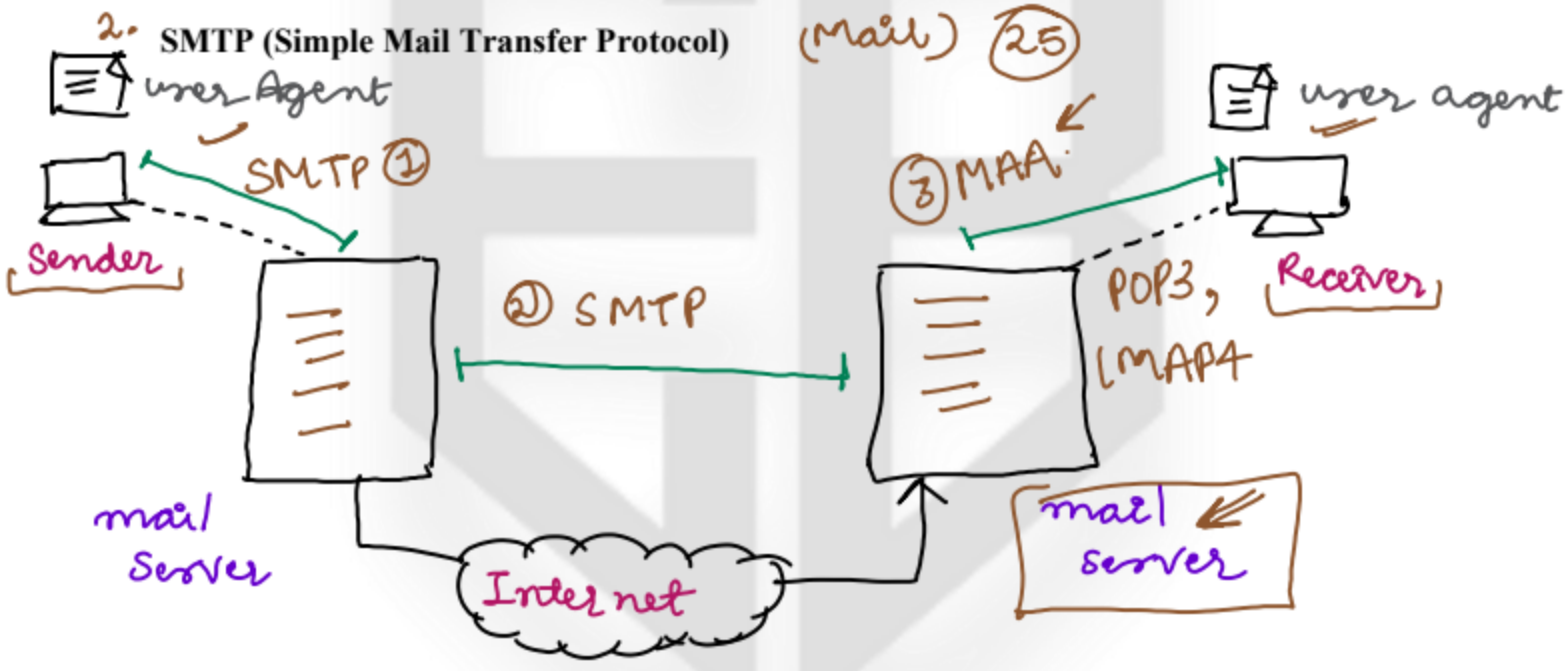
ADVANTAGES OF TELNET

+ Access remote computer.
+ Fast (No GUI)
+ Bidirectional
+ Cross-Platform (Windows, Mac, Linux).

DISADVANTAGES OF TELNET

- Not Secure (Clear-Text)
- It is Old

2. **SMTP (Simple Mail Transfer Protocol)**



1) UA Read,
2) MTA
3) MAA Message access agent

User Agent:

A user agent is a Package "or in simple words a program" of a software that composes, Reads, Responds to, and forward messages. It also handles user computers with local mailboxes.

Mail Transfer Agent "MTA":

The actual mail transmission is done through MTAs. A system must have the client MTA for sending mail, and a system must have a server MTA for receiving mail.

Message Access Agent "MAA":

The client must use the server to pull messages that use an agent for accessing messages. There are currently two protocols for accessing messages:

Post Office Protocol version 3 "POP3" and Internet Mail Access Protocol "IMAP".

POP3 protocol

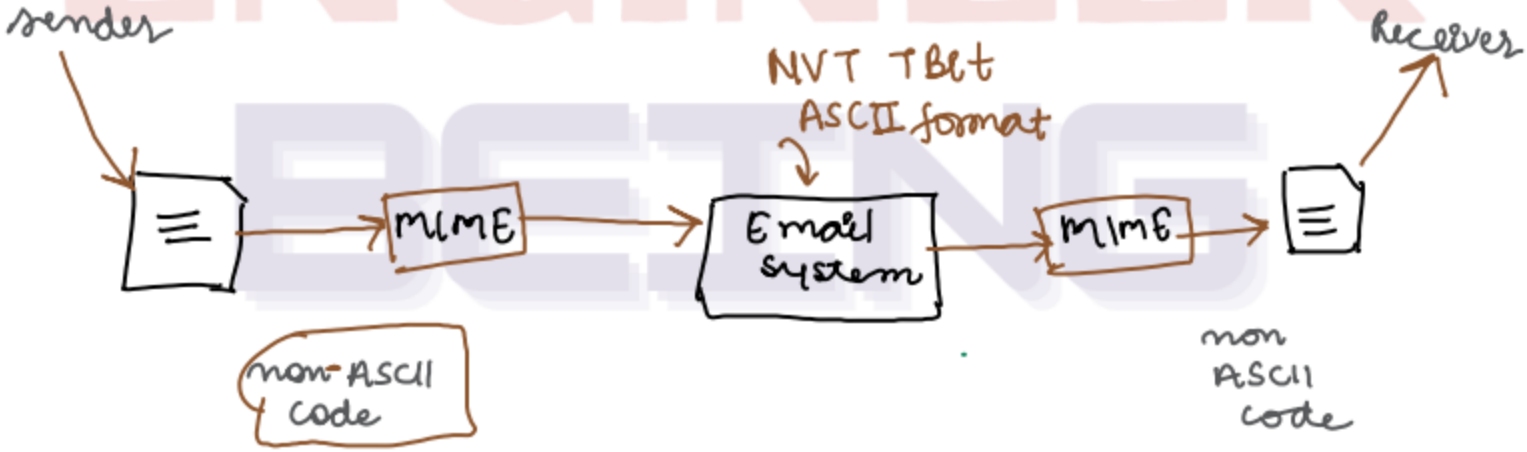
- Post Office Protocol, version 3 (POP3) is simple and limited in functionality.
- Mail access starts with the client when the user needs to download e-mail from the mailbox on the mail server.
- The client opens a connection to the server on TCP port 110.
- It then sends its user name and password to access the mailbox. The user can then list and retrieve the mail messages, one by one .

IMAP Protocol

- Another mail access protocol is Internet Mail Access Protocol, version 4 (IMAP4).
- IMAP4 is similar to POP3, but it has more features; IMAP4 is more powerful and more complex.
- A user can check the e-mail header prior to downloading.
- A user can search the contents of the e-mail for a specific string of characters prior to downloading.
- A user can partially download e-mail. This is especially useful if bandwidth is limited and the e-mail contains multimedia with high bandwidth requirements.
- A user can create, delete, or rename mailboxes on the mail server.
- A user can create a hierarchy of mailboxes in a folder for e-mail storage.

MIME(Multipurpose Internet Mail Extension)

It is a supplementary protocol that allows non-ASCII data to be sent through e-mails. (NVT 7-BIT ASCII format)



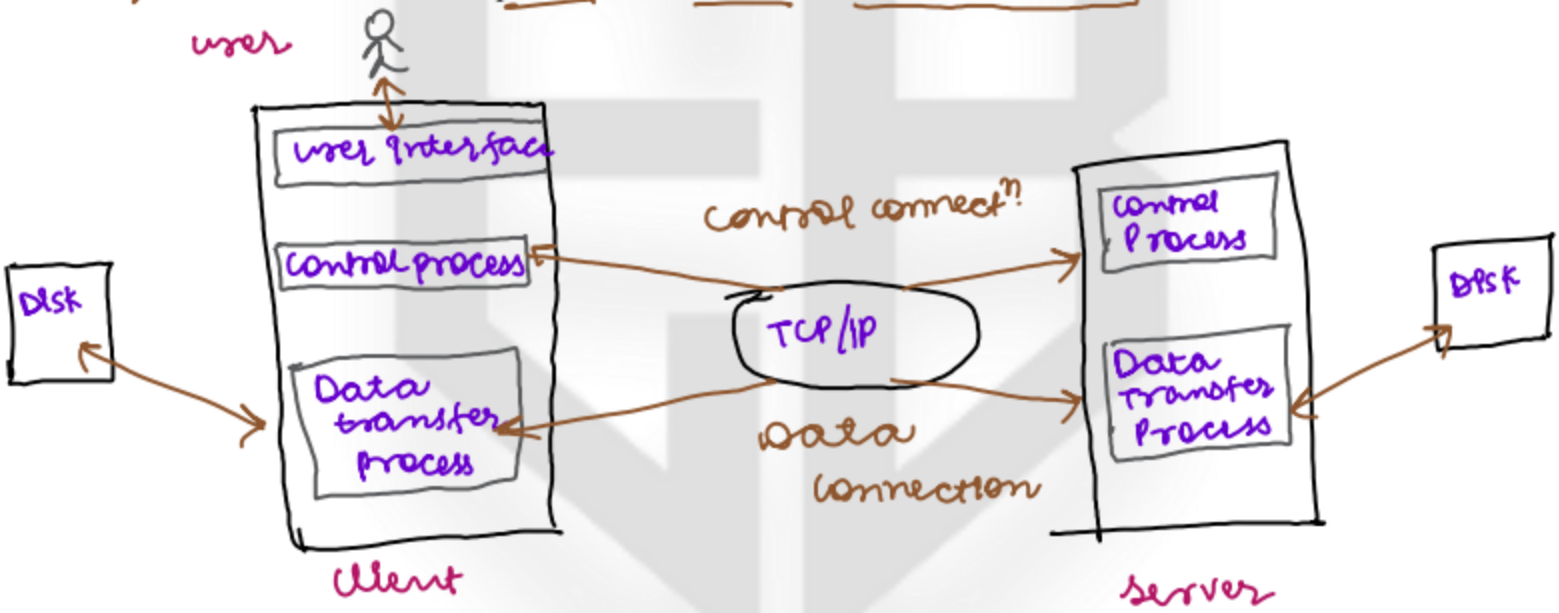
ENGINEER BEING

3 FILE TRANSFER PROTOCOL (FTP)

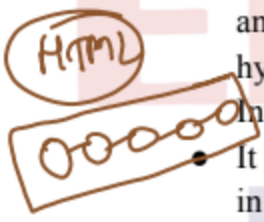


FTP uses the services of TCP.
It needs two TCP connections (1) 21
The well-known port 21 is used for the **control connection** and the well-known port 20 for the **data connection.** (2) 20

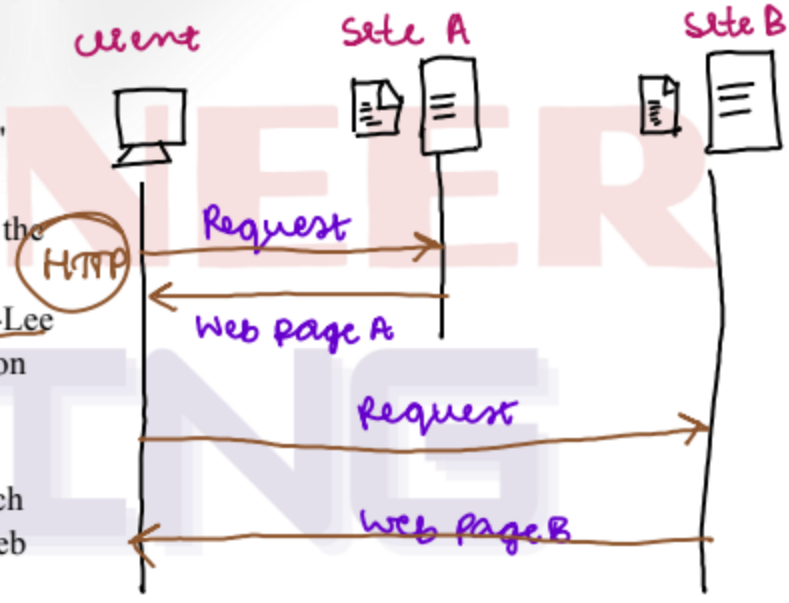
- The control connection remains connected during the entire interactive FTP session.
- The data connection is opened and then closed for each file transferred.



4 WWW(World Wide Web)

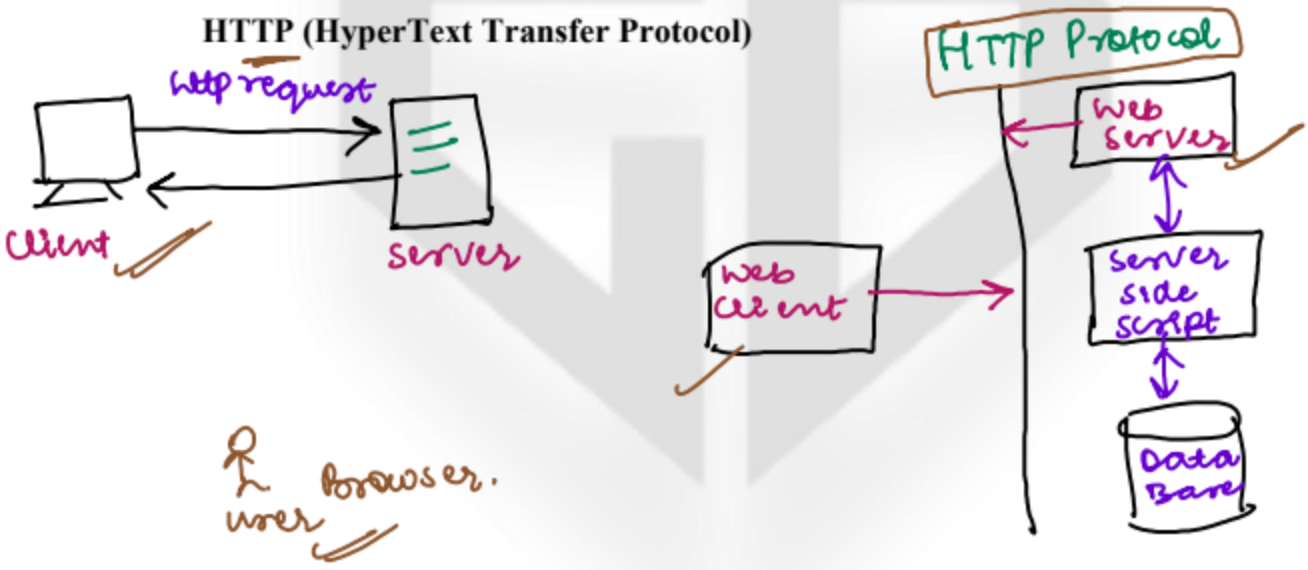


- www stands for "World Wide Web" and is a system of interconnected hypertext documents accessible via the Internet.
- It was invented by Sir Tim Berners-Lee in 1989 as a way to share information globally using hyperlinks.
- The www is based on the HTTP (Hypertext Transfer Protocol), which allows users to navigate between web pages by clicking on links.



- Web addresses, or URLs (Uniform Resource Locators), are used to locate specific web pages or resources on the www.
- Web pages are written in HTML (Hypertext Markup Language), which defines the structure and content of a page using tags.
- The www enables the display of multimedia content, including text, images, audio, and video.
- Search engines, such as Google, Bing, and Yahoo, index web pages to make them searchable and provide relevant results to users.
- Web browsers, like Chrome, Firefox, and Safari, allow users to access and view web pages by rendering HTML and executing client-side scripts.

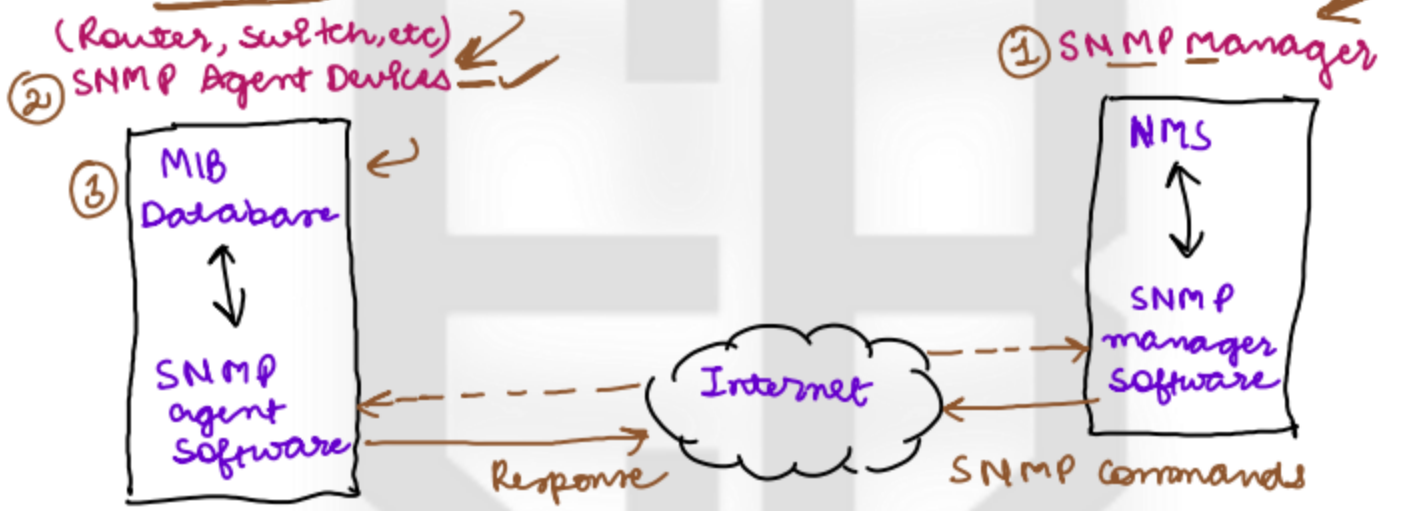
HTTP (HyperText Transfer Protocol)



- HTTP is a request/response protocol which is based on client/server based architecture. In this protocol, web browser, search engines, etc. behave as HTTP clients and the Web server like Servlet behaves as a server
- It is the protocol that allows web servers and browsers to exchange data over the web.
- It is a request response protocol.
- It uses the reliable TCP connections by default on TCP port 80.
- It is stateless means each request is considered as the new request. In other words, server doesn't recognize the user by default.

Simple Network Management Protocol(SNMP)

SNMP is an application layer protocol that uses UDP port number 161/162. SNMP is used to monitor the network, detect network faults, and sometimes even used to configure remote devices.



SNMP components –

1. **SNMP Manager:** The SNMP manager is the central system responsible for monitoring and managing the network devices. It collects and analyzes information from the managed devices and can send commands or configuration changes to them. The SNMP manager communicates with the SNMP agents on the managed devices using SNMP messages.

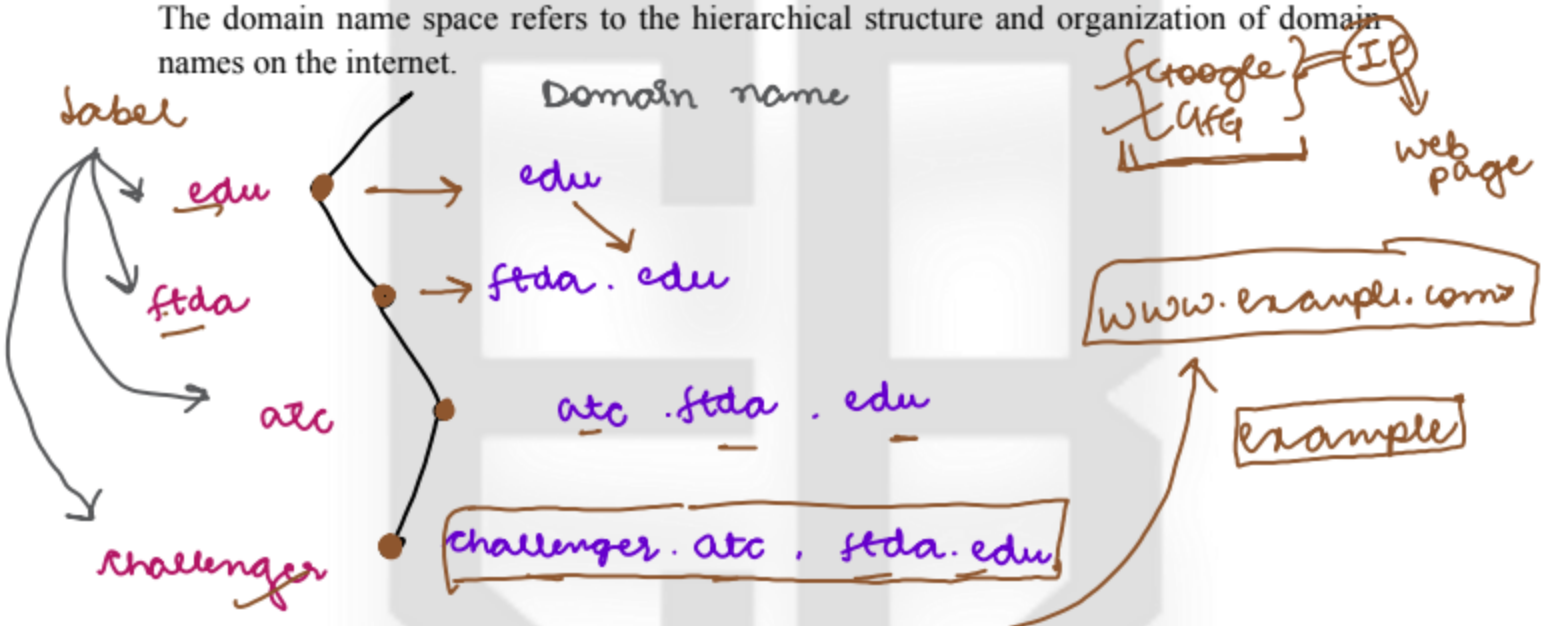
2. **SNMP Agent:** The SNMP agent is a software module residing on each managed device. Its primary role is to gather and store information about the device's configuration, status, and performance. The agent exposes this information in a structured manner, making it accessible to the SNMP manager.

3. **MIB (Management Information Base)** is a structured database used in SNMP (Simple Network Management Protocol) to define and organize the information about network devices. It enables SNMP managers to retrieve and control data from SNMP agents on managed devices.

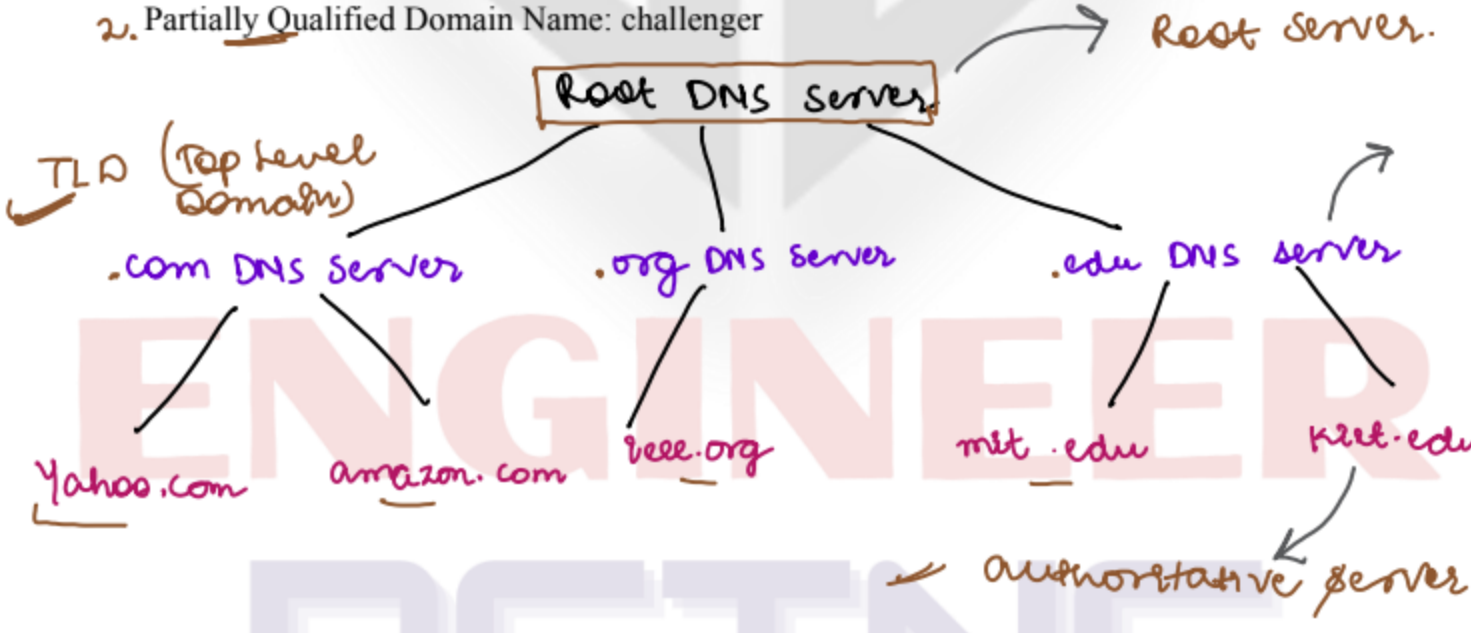
SNMP messages include GET (retrieve information), SET (configure devices), and TRAP (notify of events)

Domain Name System

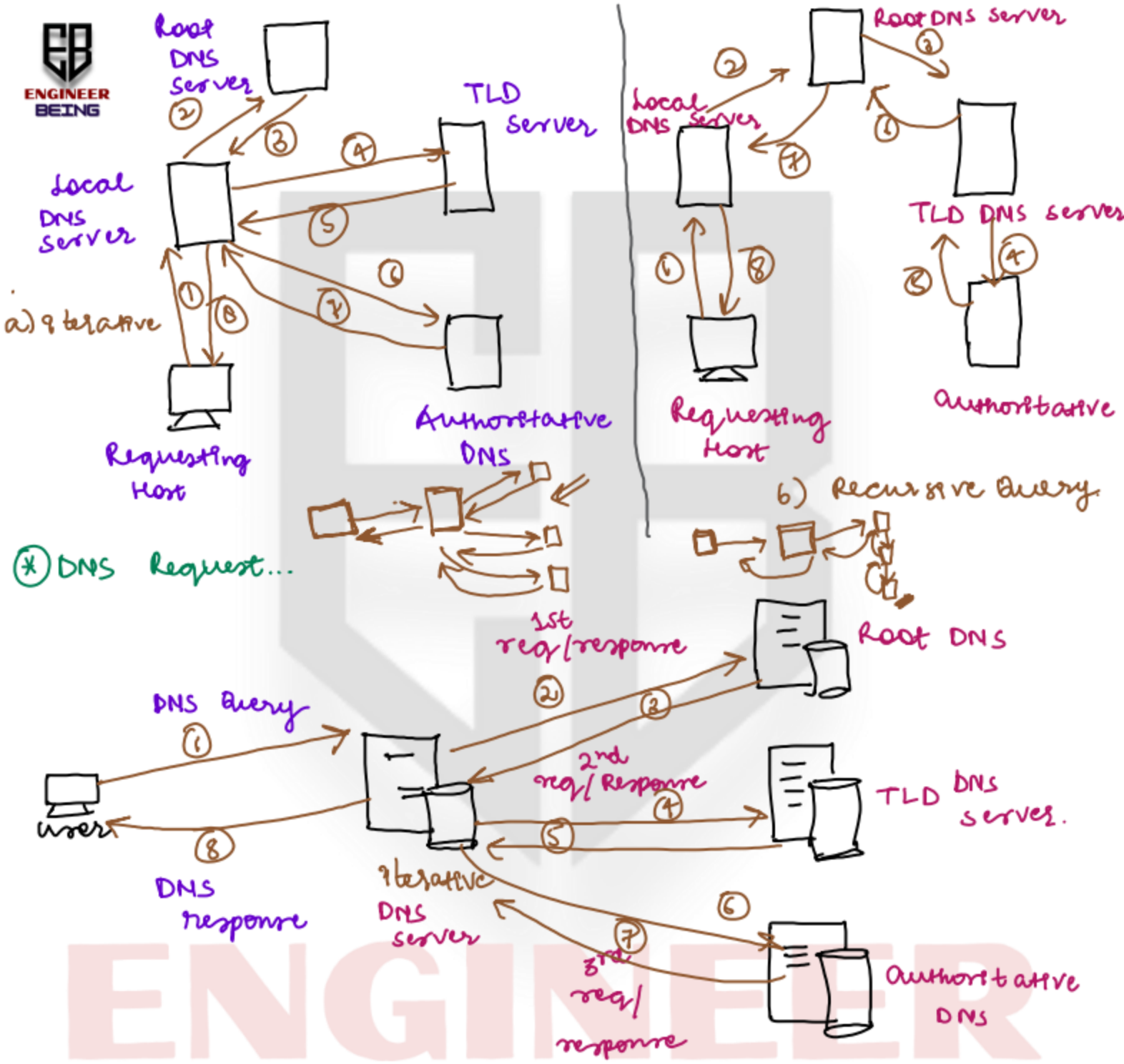
The domain name space refers to the hierarchical structure and organization of domain names on the internet.



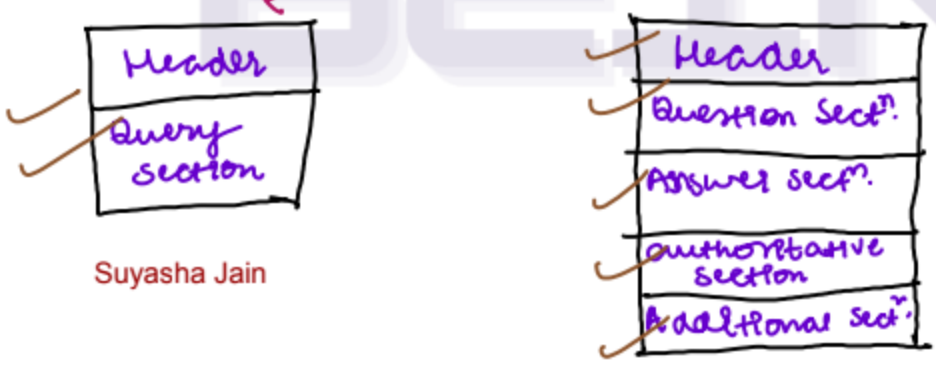
1. Fully Qualified Domain Name: challenger.ate.tbda.edu.
2. Partially Qualified Domain Name: challenger



1. Resolution in DNS: process of translating IP addresses to domain names.
   - In Iterative DNS resolution the client communicates directly with each DNS server involved in the lookup.
   - A recursive DNS lookup is where one DNS server communicates with several other DNS servers to hunt down an IP address and return it to the client.



DNS MESSAGES: Query and response



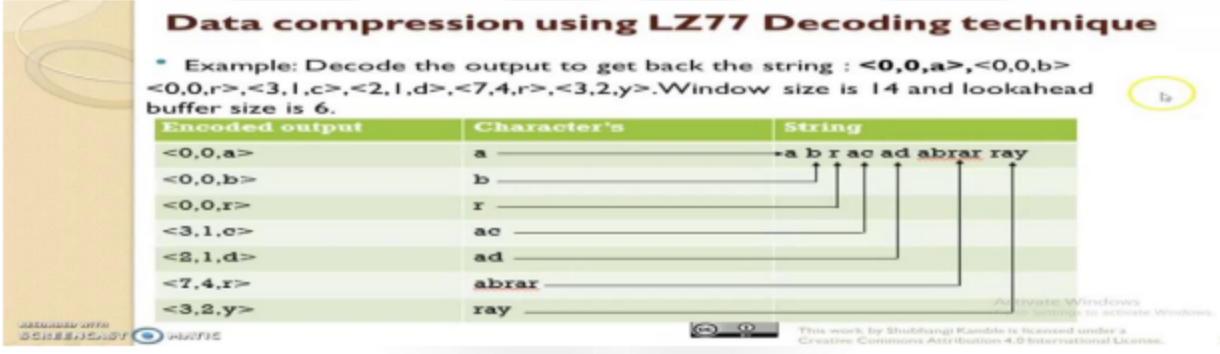
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DATA COMPRESSION

- Data compression is a reduction in the number of bits needed to represent data.
- Compressing data can save storage capacity, speed up file transfer and decrease costs for storage hardware and network bandwidth.
- Types
- Lossless-In lossless compression, the data is transformed through encoding, and its size is reduced
- Lossy-In lossy, the insignificant piece of data is removed to reduce the size.

Joint Photographic Experts Group (JPEG) is used to compress images.

- Moving Picture Experts Group (MPEG) is used to compress video.



CRYPTOGRAPHY

Cryptography is the study of secure communications techniques that allow only the sender and intended recipient of a message to view its contents

- PT(Plain Text)
- CT(Cipher Text)
- Encryption
- Decryption

Private key cryptography, also known as symmetric-key cryptography, is a cryptographic system where the same key is used for both encryption and decryption.

Public key cryptography, also known as asymmetric-key cryptography, is a cryptographic system that uses a pair of mathematically related keys: a public key and a private key.

RSA public key cryptography

- selecting two large primes at random: p, q
- computing their system modulus n=p.q

  note ø(n)=(p-1)(q-1)
- solve following equation to find decryption key d

  $e.d=1 \bmod ø(n)$ and $0 \leq d \leq n$
- C = M^e mod n
- M = C^d mod n

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DNS MESSAGES: Query and response

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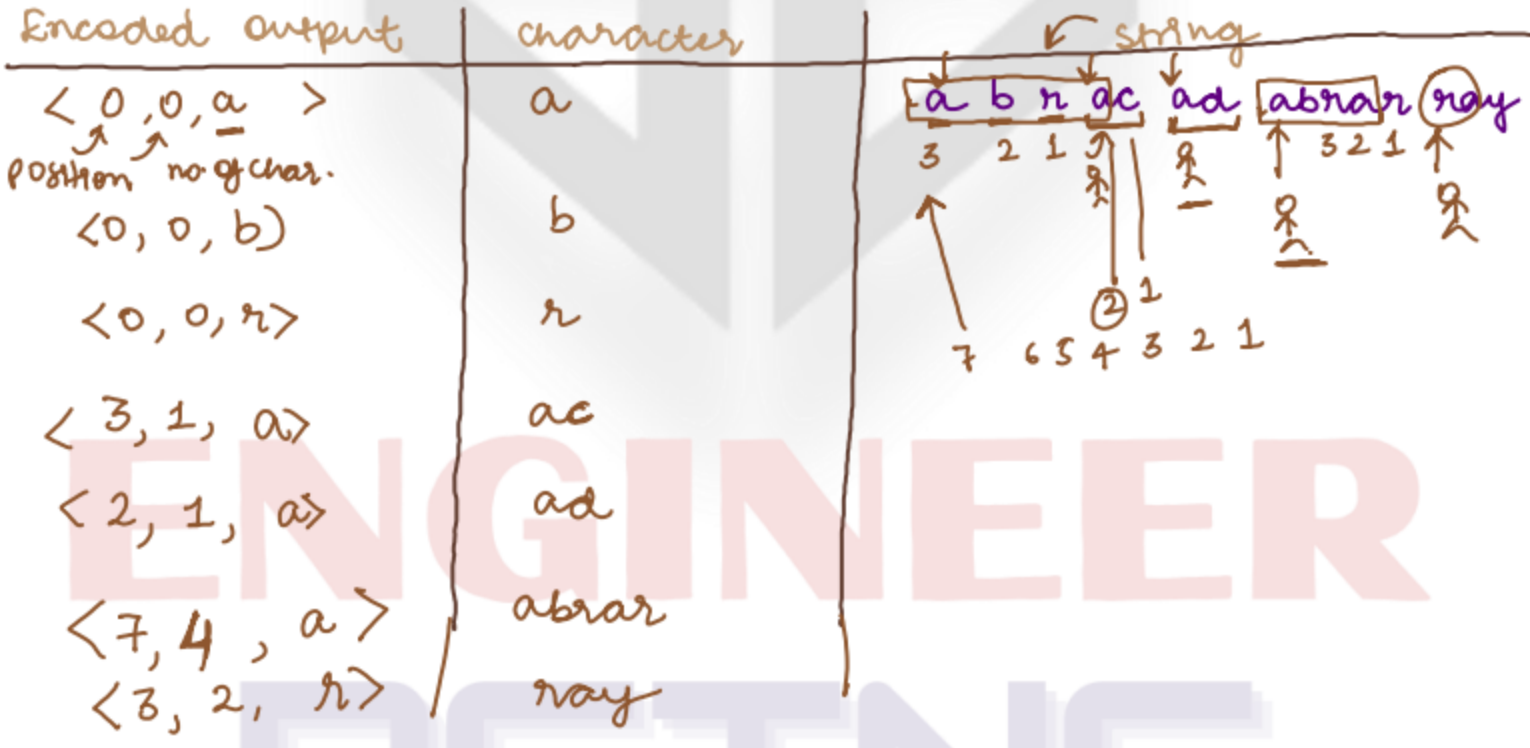
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Data compression using LZ77 Decoding techniques-



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  note ø(n)=(p-1)(q-1)
- solve following equation to find decryption key d
  ~~e.d=1 mod ø(n) and 0≤d≤n~~ $(d \cdot e) \% \phi(n) = 1$
- C = M^e mod n
- M = C^d mod n

$n = 3 \times 11 = 33$

$\phi(n) = 2 \times 10 = 20$

$7 \% 20 = 7$ d=1
$14 \% 20 = 14$ d=2
$21 \% 20 = 1$

$(d \cdot 7) \% 20 = 1$ ← d=3 ★

Ex- p=3 q=11 e=7 M=2

encrypn $C = 2^7 \% 33 = 2^7 \% 33 = 29$

decrypn $m = 29^3 \% 33 = 2$

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